DO NOW — RECALL

Lesson 093 · Von Neumann and Harvard Architectures

Name: _____ Class: _____ Date: _____

1. What does an address bus carry?

2. What does a data bus transfer?

3. Define contention.

4. A D flip-flop has D=1 and Q=0 immediately before a rising edge. What is Q just after?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Memory location.

2. Model answer

Content value/instruction bits.

3. Model answer

Two operations compete for same limited resource.

4. Model answer

Q becomes1, sampled at that active edge.

By the end, you can…

• Von Neumann stores programs and their data in a common memory/address space, so instruction fetches and data transfers may contend for shared resources.

• Harvard architecture separates instruction and data pathways, potentially allowing simultaneous instruction fetch and data access.

• Modern systems often use unified main memory but separate instruction/data caches; the classification is a model of design trade-offs, not a universal speed ranking.

Vocabulary — know these terms

Von Neumann — Shared instruction/data memory organisation.

Harvard — Separate instruction/data stores or pathways.

bottleneck — Limiting shared resource.

modified Harvard — Modern mixed memory hierarchy.

NEW LEARNING

Read, annotate and explain

Observe

Von Neumann stores programs and their data in a common memory/address space, so instruction fetches and data transfers may contend for shared resources.

Explain

Harvard architecture separates instruction and data pathways, potentially allowing simultaneous instruction fetch and data access.

Evaluate

Modern systems often use unified main memory but separate instruction/data caches; the classification is a model of design trade-offs, not a universal speed ranking.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Explain one potential bottleneck of simple Von Neumann design.

Explain the mechanism

Instruction and data transfers share memory/bus resources, so competing accesses can wait.

Describe the shared resource and its effect, not claim all Von Neumann machines are slow.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State difference between Von Neumann and Harvard memory organisation. [4]

Question 2

Explain Von Neumann bottleneck with instruction fetch and data read. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate 'Harvard architecture always makes programs faster than Von Neumann' with contemporary caches and workload. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Contemporary processors can combine a unified main memory with separate instruction and data caches.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.